Glossary: 04VirtualWork

English	中文	Brief explanation
Virtual displacement	虚位移	A hypothetical infinitesimal displacement $\delta\vec{r}$ consistent with the constraints of the system at a fixed instant of time, obtained by setting $\delta t = 0$.
Physical displacement	实位移	The actual infinitesimal displacement $d\vec{r}$ occurring over a time interval dt, including the explicit time dependence through $\partial\vec{r}/\partial t$.
Virtual work	虚功	The work $\delta W = \vec{F} \cdot \delta\vec{r}$ done by a force along a virtual displacement; in generalized coordinates $\delta W = \sum_i Q_i \, \delta q_i$.
Generalized force	广义力	Defined as $Q_i = \vec{F} \cdot (\partial\vec{r}/\partial q_i)$; the component of force conjugate to the generalized coordinate q_i.
Constraint force	约束力	The force arising from constraints, written in generalized coordinates as $Q_i^c = \lambda \, \partial g/\partial q_i$ for a holonomic constraint $g(q_i, t) = 0$.
Applied force	主动力	The external or active force directly imposed on the system, as distinguished from constraint forces.
Ideal constraint	理想约束	A constraint for which the total virtual work done by constraint forces vanishes for all allowed virtual displacements.
Principle of virtual work	虚功原理	In static equilibrium with ideal constraints, the total virtual work done by the applied forces is zero: $\sum_i Q_i \, \delta q_i = 0$.
Holonomic constraint	完整约束	A constraint expressible as an equation $g(q_i, t) = 0$ relating the generalized coordinates and possibly time.
Lagrange multiplier	拉格朗日乘	An auxiliary parameter $\lambda(t)$ (or $\lambda_j(t)$ for multiple constraints $g_j = 0$) that enforces the constraint and determines the magnitude of the constraint force.

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Degrees of freedom	自由度	The number of independent coordinates that can be varied in a small displacement of the system.
Independent coordinates	独立坐标	Generalized coordinates whose virtual variations δq_i can be chosen arbitrarily; the vanishing of their coefficients gives the equilibrium conditions.
Static equilibrium	静力平衡	A state in which the system experiences no acceleration; all generalized forces vanish ($Q_i = 0$) or, for conservative systems, $\partial V/\partial q_i = 0$.
Equilibrium condition	平衡条件	The requirement that each generalized force be independently zero, $Q_i = 0$, derived from the independence of the virtual displacements.
Conservative system	保守系统	A system in which all applied forces derive from a scalar potential energy V, so that $Q_i = -\partial V/\partial q_i$.
Potential energy	势能	The scalar function $V(q)$ whose partial derivatives give the generalized forces of a conservative system.
Stationary point	驻点	A configuration where the first derivatives of the potential energy vanish, $\partial V/\partial q_i = 0$, yielding a possible static equilibrium.
d'Alembert's principle	达朗贝尔原理	The extension of the principle of virtual work to dynamics: an accelerating system is treated as being in equilibrium in a non-inertial frame co-moving with the system, provided fictitious forces $-m\ddot{\vec{r}}$ are included.
Fictitious force	虚构力 / 惯性力	The force $-m\vec{a}$ (or $-m\ddot{\vec{r}}$) introduced in a non-inertial frame to enable the use of static equilibrium methods for dynamical problems.
Non-inertial frame	非惯性参	A reference frame that accelerates relative to an inertial frame; fictitious forces must be added to apply equilibrium conditions.

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Lagrange's equation	拉格朗日方程	Derived from d'Alembert's principle: $\frac{d}{dt}\left(\partial T/\partial \dot{q}_i\right) - \partial T/\partial q_i = Q_i$, valid for arbitrary generalized forces.
Euler–Lagrange equation	欧拉–拉格朗日方程	For conservative systems, $\frac{d}{dt}\left(\partial L/\partial \dot{q}_i\right) - \partial L/\partial q_i = 0$, where $L = T - V$ is the Lagrangian.
Non-conservative force	非保守力	Forces not derivable from a potential; they appear as Q_i^{nc} on the right-hand side of the Euler–Lagrange equation.
Kinetic energy	动能	$T = \frac{1}{2}m\dot{\mathbf{r}}^2$ in Cartesian coordinates; in generalized coordinates its derivatives with respect to $\dot{q}_i$ and q_i produce the inertial terms in Lagrange's equation.